

ECE 3337: Signals & Systems Analysis

Lecture 9: Linear Time Invariant Systems

**Reference: Chapter 2
(Section 2-8)**

Please turn off your gadgets now



Exam 1 This Saturday

Syllabus: Chapters 1 and 2, as covered in class

Date and Time: Saturday, Sept 27, 10 am – 11:30 am

(Starts at 9.30 am for students with DART center accommodations)

Location: W122 Engineering Building (both sections)

Help Sessions: Fridays, 12.30 pm – 2 pm

Old Exams with solutions are on the Teams site for practice

1. Try the old exams without looking at the answers
2. Read the assigned sections to understand the basic concepts and memorize them.
3. Attempt the problems and practice until you can do them fully on your own without notes, like in a test.
4. Practice for correctness and speed

Math Moment: Quadratic Equations

$$s^2 + a_1s + a_2 = 0$$

$$(s - p_1)(s - p_2) = 0$$

$$s^2 - (p_1 + p_2)s + p_1p_2 = 0$$

Roots

$$p_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}$$

$$p_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$

$a_1^2 > 4a_2$ ☐ Both roots are real

$a_1^2 < 4a_2$ ☐ Complex conjugate roots

$a_1^2 = 4a_2$ ☐ Identical roots

$$p_1 + p_2 = -a_1$$

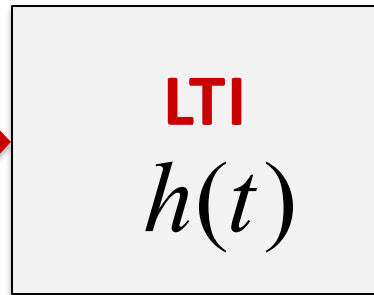
$$p_1p_2 = a_2$$



Recap: LTI Response to Sinusoids

SINUSOID IN

$$x(t) = A \cos(\omega t + \phi)$$



$$y(t) = A \underbrace{|\hat{\mathbf{H}}(\omega)|}_{\text{Amplitude Change}} \cos(\omega t + \phi + \underbrace{\theta}_{\text{Phase Change}})$$

SAME FREQUENCY
SINUSOID OUT

Amplitude
Change

Phase
Change

$$\theta = \angle \hat{\mathbf{H}}(\omega)$$

EQUIVALENT
REPRESENTATIONS



$$\hat{\mathbf{H}}(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$



Example

Calculate the response of the same LTI system to the following new input:

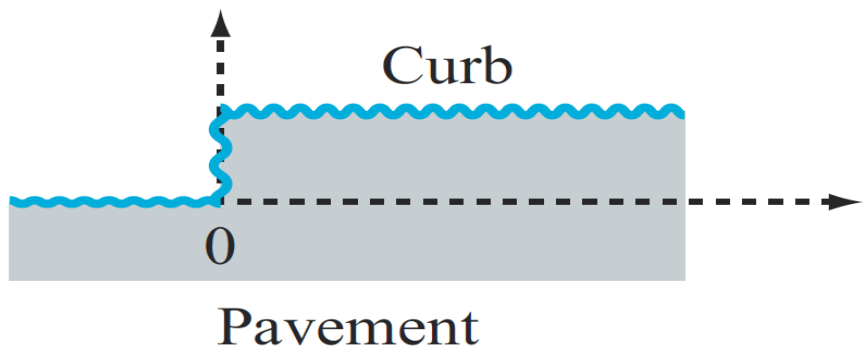
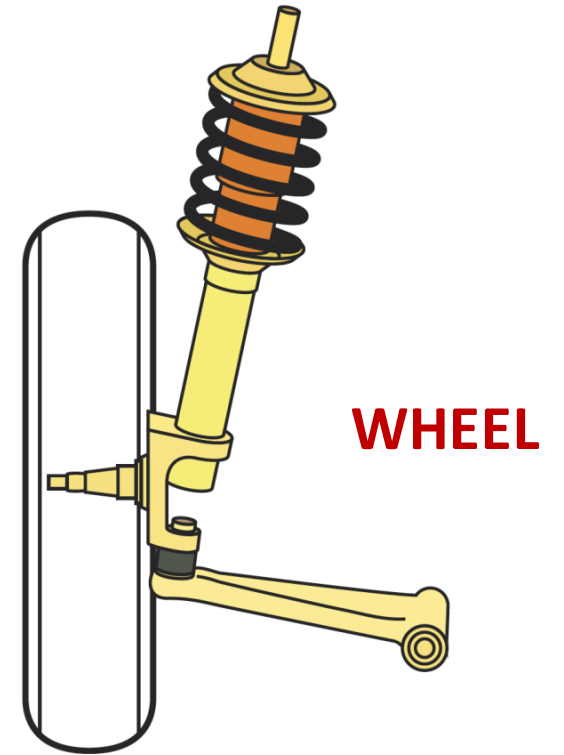
$$x(t) = 7 + \cos(2t)$$

Correct Answer:

$$y(t) = 0 + 2\cos(2t - 45^\circ)$$



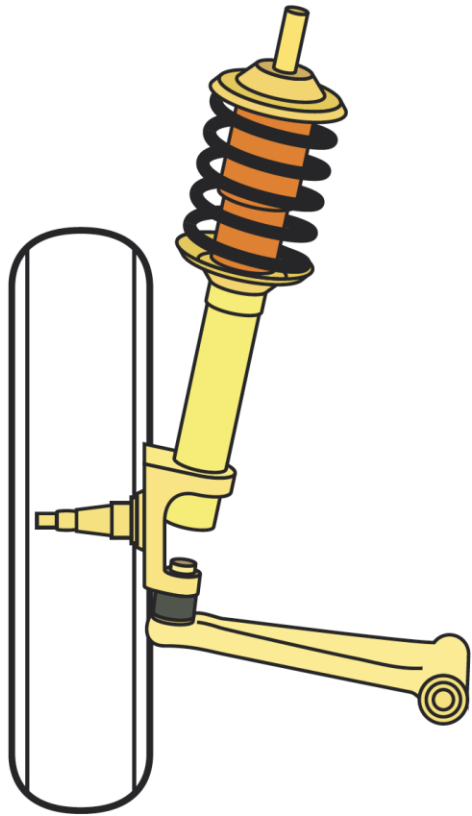
Today: Car Hits Curb (step input)





Today: Car Hits Curb

DIFFERENTIAL EQUATION DESCRIBING THE CAR'S FRONT WHEEL SUSPENSION



$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = b_1 \frac{dx}{dt} + b_2 x(t)$$

This LCCDE shows up in a variety of real-world LTI systems, so let us study it!



Today's Big Questions

$$\frac{d^2y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = b_1 \frac{dx}{dt} + b_2 x(t)$$

By varying the parameters a_1, a_2, b_1, b_2 we can get a variety of response patterns.

Question 1: How do the values of the parameters affect the nature of the impulse and step responses?

Question 2: How do these parameters affect the BIBO stability of the system?

Question 3: How can we optimize these parameters to build well-behaved systems?

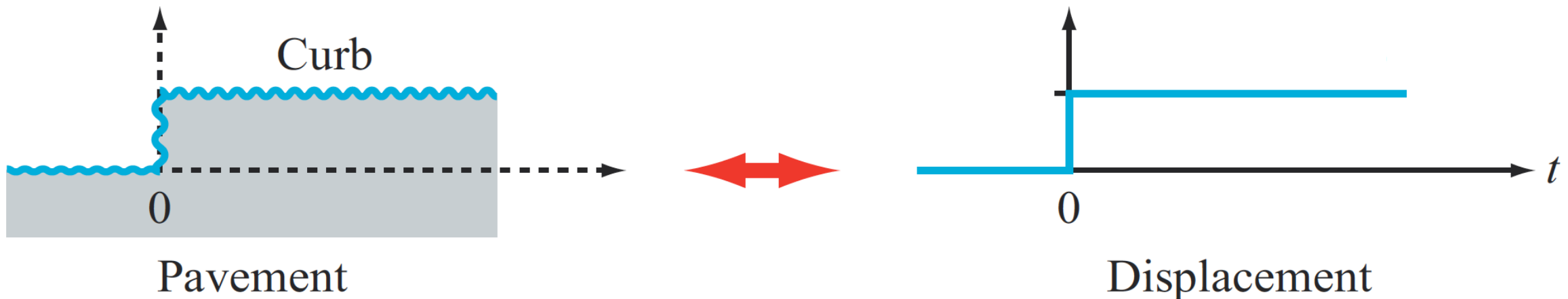


Impulse Response of 2nd Order Systems

Strategy: Start by deriving the impulse response of the LCCDE

$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = b_1 \frac{dx}{dt} + b_2 x(t)$$

Once we have the impulse response $h(t)$, we can derive the response to the step input representing the curb





Recap: Standard Format

Coefficient of highest
derivative = 1



$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = b_1 \frac{dx}{dt} + b_2 x(t)$$

Output variable on LHS

Input variable on RHS



Let's Start Simple

Consider the simplified LCCDE with no input derivatives

$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = \cancel{b_1 \frac{dx}{dt}} + b_2 x(t)$$

$$b_2 = 1$$

$$b_1 = 0$$

So, it simplifies to:

$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = x(t)$$



Solving the Differential Equation

$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = x(t)$$

Assume that $y(t)$ has a solution of the form:

$$y(t) = Ae^{st}$$

$$\frac{dy}{dt} = Ase^{st} = sy(t)$$

$$\frac{d^2 y}{dt^2} = As^2 e^{st} = s^2 y(t)$$

$$\square \quad s^2 y(t) + a_1 sy(t) + a_2 y(t) = x(t)$$



LCCDE with no Input Derivative

We need to solve :

$$s^2 y(t) + a_1 s y(t) + a_2 y(t) = x(t)$$

First step: Let us set $x(t) = 0$ (homogeneous form)

☐ $(s^2 + a_1 s + a_2) y(t) = 0$

☐ $s^2 + a_1 s + a_2 = 0$

$$(s - p_1)(s - p_2) = 0$$

**THE CHARACTERISTIC EQUATION
DESCRIBES THE LTI SYSTEM'S
“ZERO INPUT” RESPONSE**

**(yes, we can get a response even when
the input is zero)**



Revisiting The RC Circuit

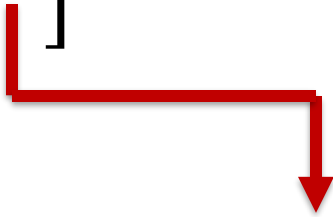
First-order Differential Equation:

$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$$

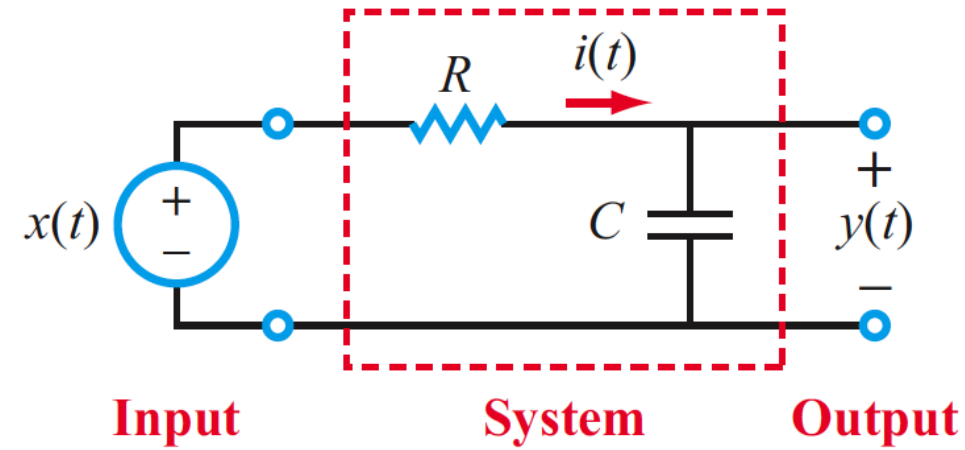
Operator Notation:

$$\left[\frac{d}{dt} + \frac{1}{RC} \right] y(t) = \frac{1}{RC} x(t)$$

$$\left[\frac{d}{dt} - p \right] y(t) = -p x(t), \text{ where } p = \frac{-1}{RC}$$


$$h(t) = -p e^{pt} u(t)$$

Impulse Response



$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

We can use this result to solve higher-order equations by factoring them into first-order terms



Factoring the Differential Equation

The differential equation can be “factored” as follows:

$$\frac{d^2y}{dt^2} - (p_1 + p_2)\frac{dy}{dt} + (p_1p_2)y = x(t)$$

$$\underbrace{\left[\frac{d}{dt} - p_1\right]}_{\text{pink bracket}} \underbrace{\left[\frac{d}{dt} - p_2\right]y(t)}_{z(t)} = x(t)$$

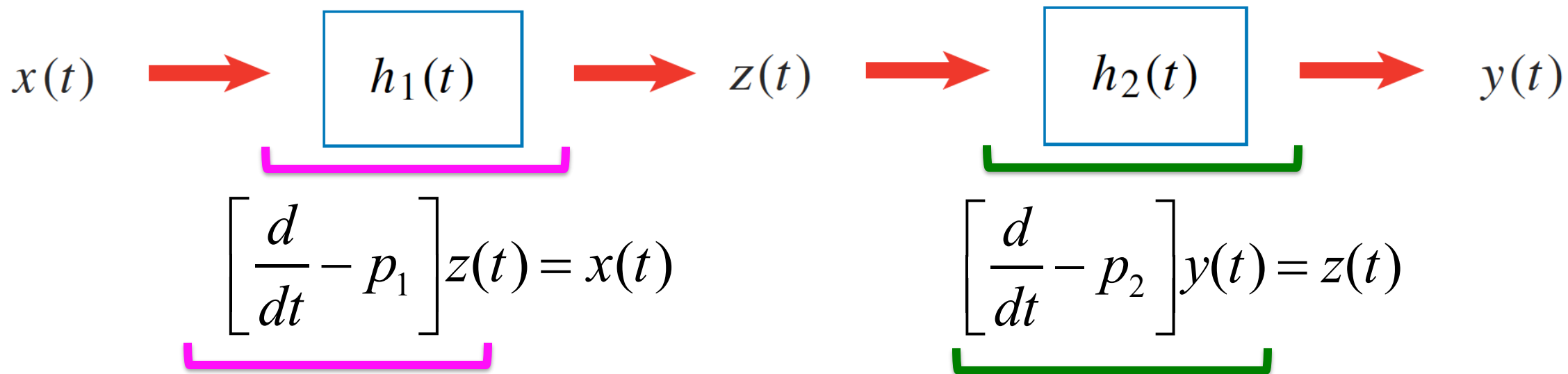
THIS CAN BE THOUGHT OF AS A CASCADE OF TWO SYSTEMS

$$\underbrace{\left[\frac{d}{dt} - p_2\right]y(t)}_{\text{green bracket}} = z(t) \quad \rightarrow \quad \underbrace{\left[\frac{d}{dt} - p_1\right]z(t)}_{\text{pink bracket}} = x(t)$$

EACH OF THESE SYSTEMS IS MATHEMATICALLY LIKE AN RC CIRCUIT!



Rewriting the Equation



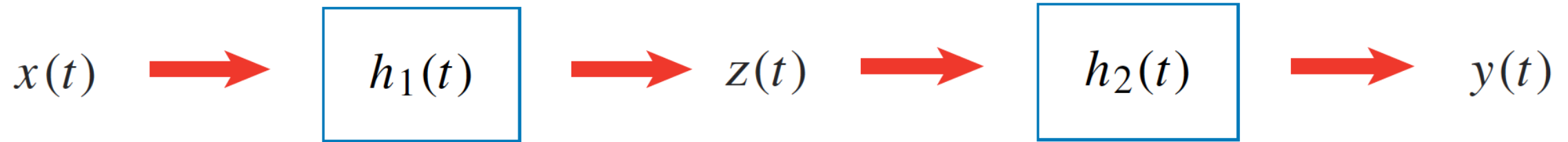
THESE FIRST-ORDER DIFFERENTIAL EQUATIONS ARE SIMILAR TO OUR RC CIRCUIT,
SO WE KNOW THEIR SOLUTIONS

$$\boxed{h_1(t) = e^{p_1 t} u(t)}$$

$$\boxed{h_2(t) = e^{p_2 t} u(t)}$$



Rewriting the Equation



$$\boxed{h_1(t) = e^{p_1 t} u(t)}$$

$$\boxed{h_2(t) = e^{p_2 t} u(t)}$$

$$h_c(t) = h_1(t) * h_2(t)$$

COMBINED RESPONSE

$$= e^{p_1 t} u(t) * e^{p_2 t} u(t)$$

$$= \frac{1}{p_1 - p_2} [e^{p_1 t} - e^{p_2 t}] u(t), \quad p_1 \neq p_2$$

**TABLE 2-2 COMES
HANDY!**



Table 2-2: Commonly encountered convolutions.

1. $u(t) * u(t) = t u(t)$

2. $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$

 **3.** $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$

4. $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$

5. $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$
 $a \neq b$

6. $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$

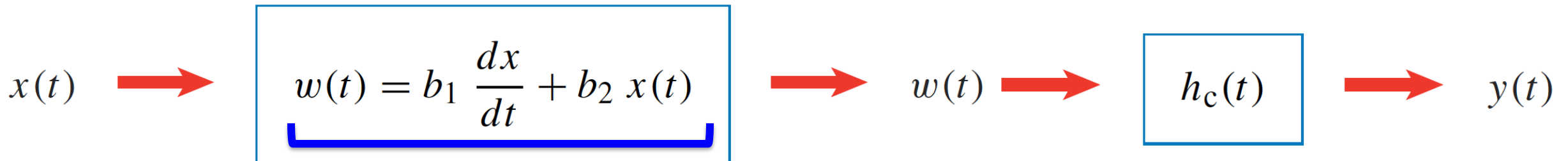


LCCDE with Input Derivative

Now, let us get back to the original full equation:

$$\frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_2 y(t) = \underbrace{b_1 \frac{dx}{dt} + b_2 x(t)}_{w(t)}$$

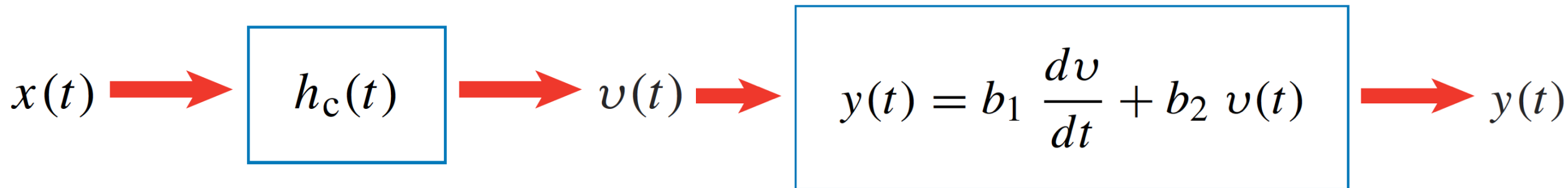
THIS CAN BE PICTURED AS FOLLOWS





Convolution is Commutative

Since convolution is commutative, we can reverse the order of the systems, as follows:



Now, if we set $x(t) = \delta(t)$, it is clear that $v(t) = h_c(t)$

This allows us to calculate $h(t)$ directly, as follows:

$$h(t) = b_1 \frac{dh_c}{dt} + b_2 h_c(t)$$



Overall Impulse Response $h(t)$

$$h(t) = b_1 \frac{dh_c}{dt} + b_2 h_c(t) \quad \text{THIS IS THE BASIS FOR CALCULATING THE RESPONSE TO ANY INPUT!}$$

$$= \left[b_1 \frac{d}{dt} + b_2 \right] \left[\frac{1}{p_1 - p_2} \right] [e^{p_1 t} - e^{p_2 t}] u(t)$$

$$= \frac{b_1 p_1 + b_2}{p_1 - p_2} e^{p_1 t} u(t) - \frac{b_1 p_2 + b_2}{p_1 - p_2} e^{p_2 t} u(t)$$

$$h(t) = A_1 e^{p_1 t} u(t) + A_2 e^{p_2 t} u(t)$$

$$A_1 = \frac{b_1 p_1 + b_2}{p_1 - p_2}$$

$$A_2 = \frac{-(b_1 p_2 + b_2)}{p_1 - p_2}$$

CONSTANTS



Observations

$$h(t) = A_1 e^{p_1 t} u(t) + A_2 e^{p_2 t} u(t)$$

$$A_1 = \frac{b_1 p_1 + b_2}{p_1 - p_2} \quad A_2 = \frac{-(b_1 p_2 + b_2)}{p_1 - p_2}$$

The roots of the characteristic equation p_1 and p_2 define the fundamental nature of the impulse response

We can infer a lot about the system by examining the roots

- **Is it damped?**
- **Is it growing without bound?**
- **Is it oscillatory?**



Guiding Examples

Describe the behaviors of the following systems:

$$h_1(t) = e^{2t}u(t) \quad \text{UNDAMPED (BIBO UNSTABLE), NOT OSCILLATORY}$$

$$h_2(t) = e^{-2t}u(t) \quad \text{DAMPED, NOT OSCILLATORY}$$

$$h_3(t) = e^{j3t}u(t) \quad \text{UNDAMPED(BIBO UNSTABLE), OSCILLATORY}$$

$$h_4(t) = e^{-2t+j3t}u(t) \quad \text{DAMPED, OSCILLATORY}$$

$$h_5(t) = e^{2t+j3t}u(t) \quad \text{UNDAMPED(BIBO UNSTABLE), OSCILLATORY}$$



Recap: Roots

The roots of the characteristic equation $s^2 + a_1s + a_2 = 0$ are given by:

$$p_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}$$

$$p_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$

- A** $a_1^2 > 4a_2$ ☐ Both roots are real
- B** $a_1^2 < 4a_2$ ☐ Complex conjugate roots
- C** $a_1^2 = 4a_2$ ☐ Identical real roots

THESE SCENARIOS DESCRIBE 3 DISTINCT SYSTEM BEHAVIORS



Preview: 3 Scenarios for the Step Response

$$y_{\text{step}}(t) = h(t) * u(t)$$

The 3 scenarios A, B, and C correspond to distinct responses

Underdamped

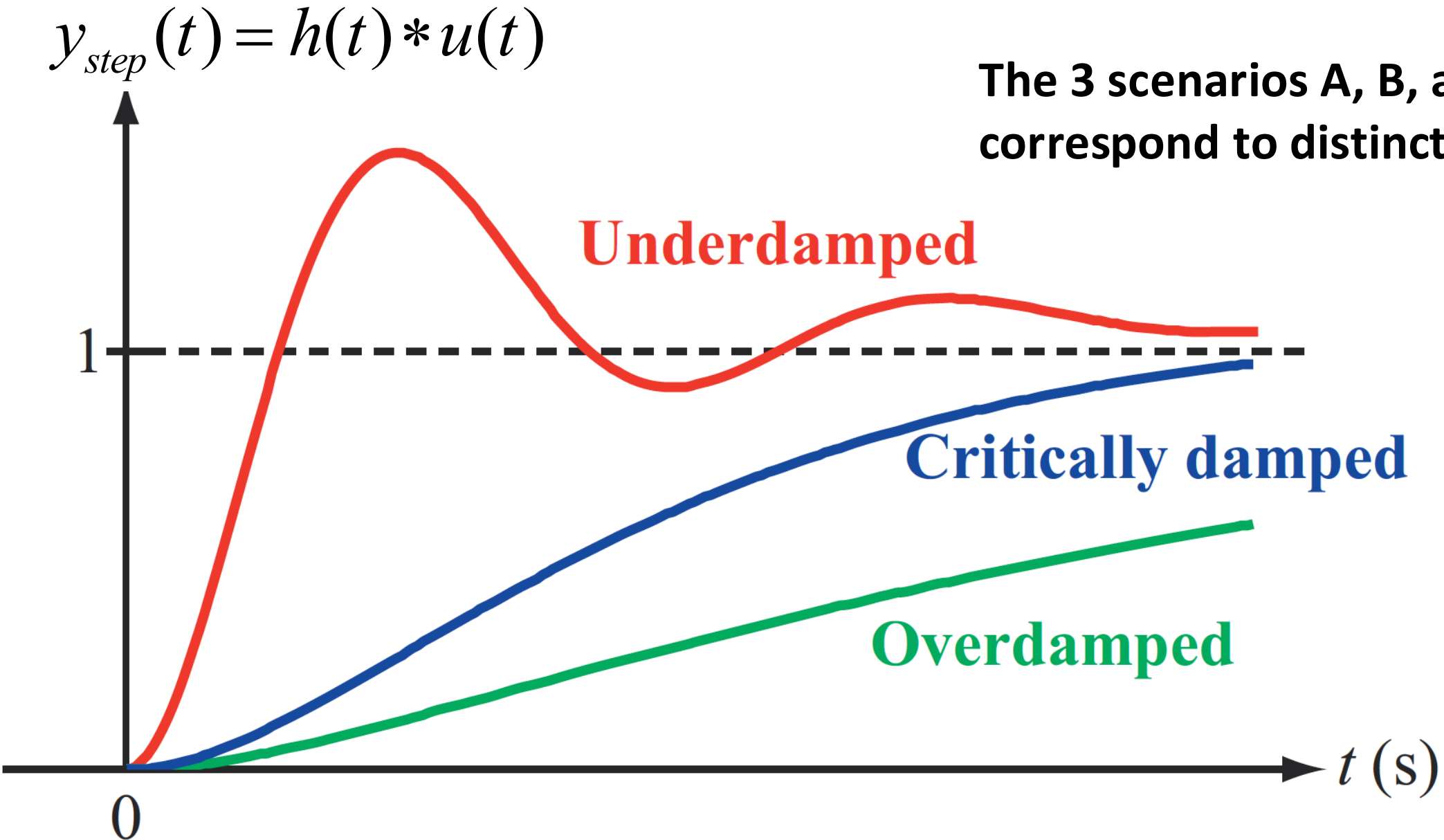
Critically damped

Overdamped

t (s)

1

0





Scenario A

DISTINCT REAL ROOTS: Overdamped Response $\xi > 1$

$$h(t) = \frac{b_1 p_1 + b_2}{p_1 - p_2} e^{p_1 t} u(t) - \frac{b_1 p_2 + b_2}{p_1 - p_2} e^{p_2 t} u(t)$$

$$= A_1 e^{p_1 t} u(t) + A_2 e^{p_2 t} u(t)$$

p_1 and p_2 must both be negative
real numbers for BIBO stability



Basic Observation

The roots of the characteristic equation $s^2 + a_1s + a_2 = 0$ are given by:

$$p_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}$$

$$p_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$

Note that $p_2 < p_1$ always

For BIBO stability, the real parts of p_1 and p_2 should be negative



Scenario A (distinct real roots)

$$p_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}$$
$$p_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$

$p_2 < p_1$, so ensuring that $p_1 < 0$ is sufficient

FOR BIBO STABILITY

$$\sqrt{a_1^2 - 4a_2} < a_1$$

We know that $\sqrt{a_1^2 - 4a_2} > 0$ for Scenario A

$$\square \quad 0 < \sqrt{a_1^2 - 4a_2} < a_1 \quad (1)$$

$$\square \quad a_1 > 0 \quad \text{CONDITION \#1}$$

Squaring (1) gives us: $a_1^2 - 4a_2 < a_1^2$

$$\square \quad 4a_2 > 0 \quad \square \quad a_2 > 0 \quad \text{CONDITION \#2}$$



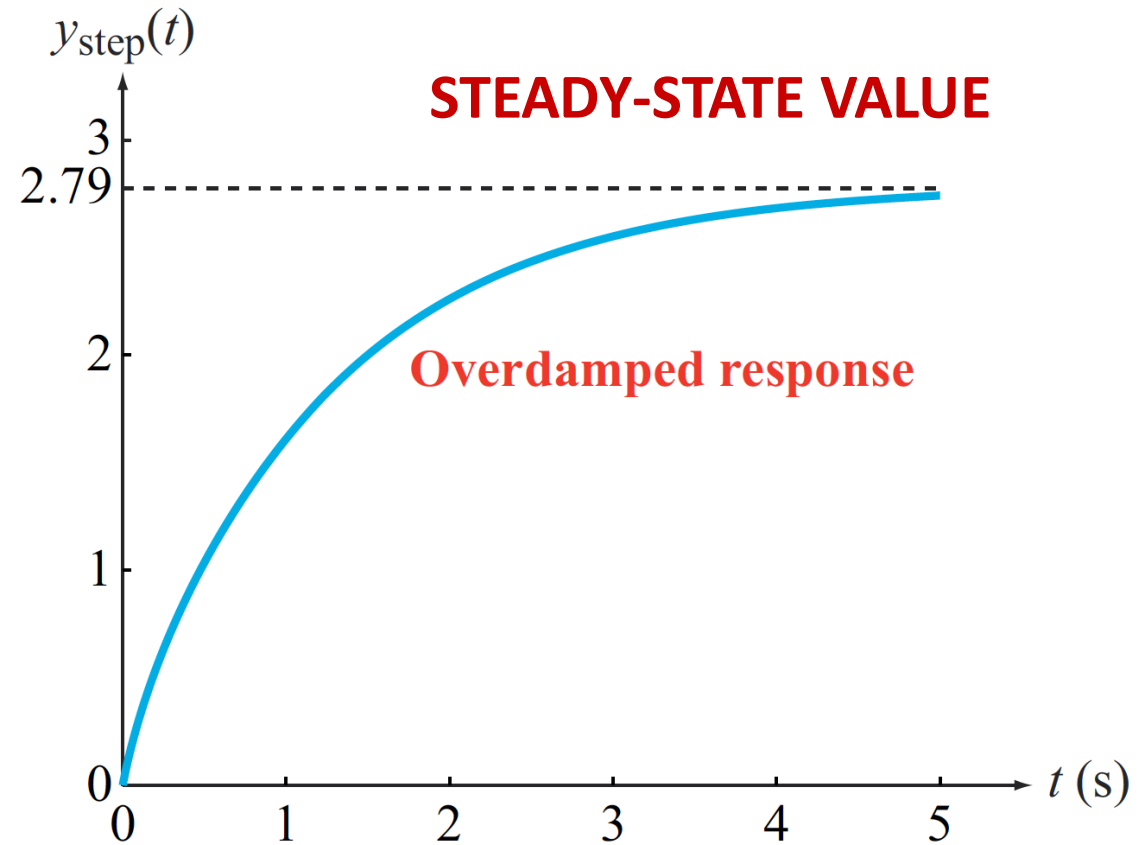
Step Response for Scenario A

DISTINCT REAL ROOTS

$$h(t) = A_1 e^{p_1 t} u(t) + A_2 e^{p_2 t} u(t)$$

$$y_{\text{step}}(t) = u(t) \int_0^t (A_1 e^{p_1 \tau} + A_2 e^{p_2 \tau}) d\tau$$

$$= \left[\frac{A_1}{p_1} (e^{p_1 t} - 1) + \frac{A_2}{p_2} (e^{p_2 t} - 1) \right] u(t)$$



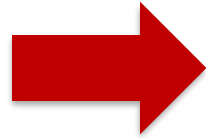
$$a_1 = 12, \quad a_2 = 9$$

$$b_1 = 4, \quad b_2 = 25$$



Some Useful Notation

$$p_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}$$
$$p_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$



$$p_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = \omega_0 \left[-\xi + \sqrt{\xi^2 - 1} \right]$$
$$p_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = \omega_0 \left[-\xi - \sqrt{\xi^2 - 1} \right]$$

$$\alpha = \frac{a_1}{2}$$

Attenuation coefficient (Np/s)

$$\omega_0 = \sqrt{a_2}$$

Undamped natural frequency (rad/s)

$$\xi = \frac{\alpha}{\omega_0}$$

Damping coefficient (dimensionless)

Np: Neper (after John Napier, inventor of the log scale)



Scenario B: Complex Conjugate Roots

$a_1^2 < 4a_2$ $\square$ Complex conjugate roots

Underdamped Response

$$p_1 = -\frac{a_1}{2} + j\omega_d \quad p_2 = -\frac{a_1}{2} - j\omega_d \quad \omega_d = \omega_0 \sqrt{1 - \xi^2} \quad \xi < 1$$

$$h(t) = A_1 e^{-\alpha t} e^{j\omega_d t} u(t) + A_2 e^{-\alpha t} e^{-j\omega_d t} u(t)$$

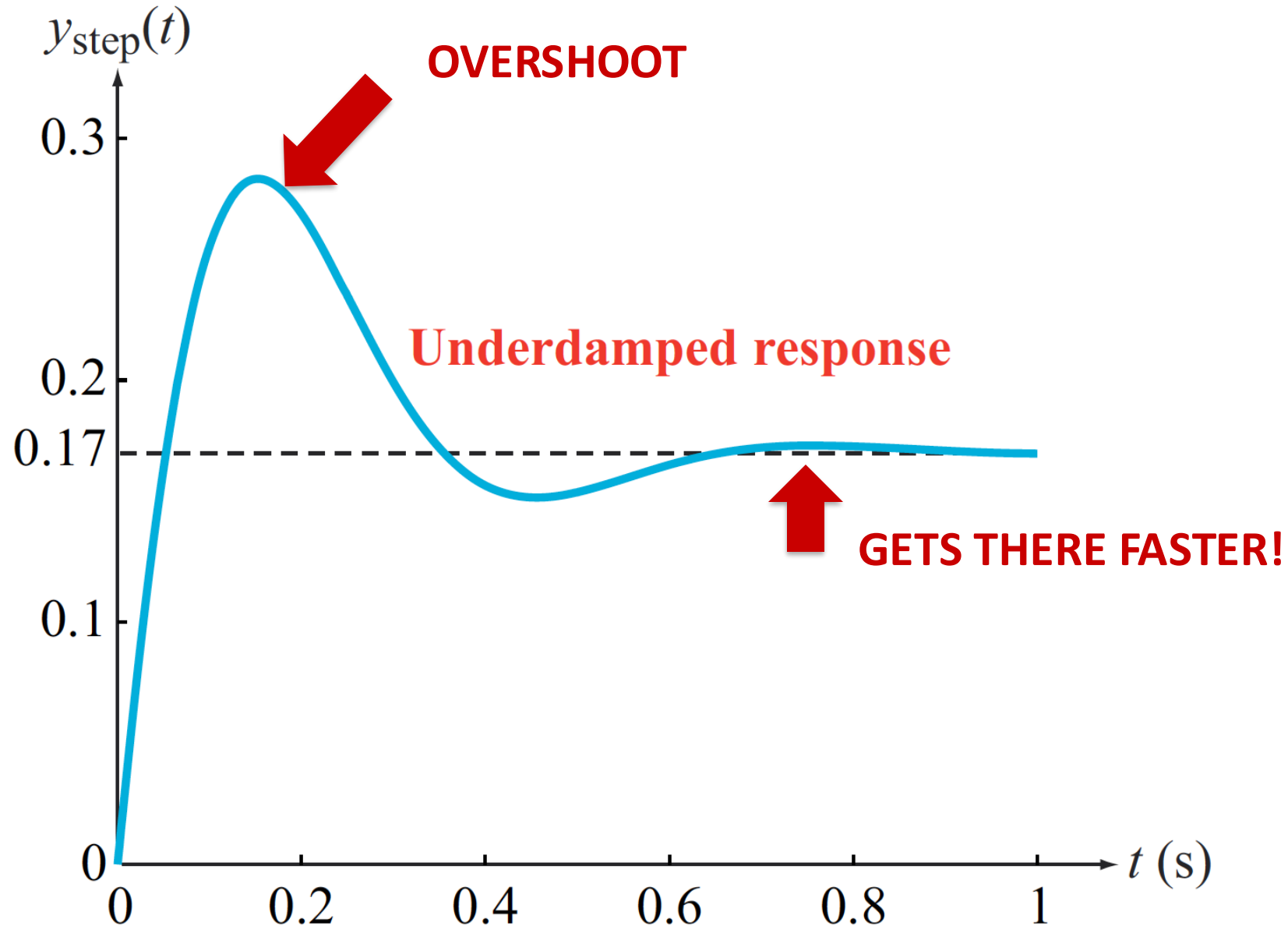
For BIBO stability, we need $\alpha = \frac{a_1}{2} > 0$ **CONDITION #1**

Since $a_1 > 0$ and $a_1^2 < 4a_2 \Rightarrow 0 < a_1^2 < 4a_2$

$\Rightarrow a_1^2 > 0$ **CONDITION #2**



Sample Step Response



$$a_1 = 12, \quad a_2 = 144$$

$$b_1 = 4, \quad b_2 = 25$$



Scenario C: Equal Roots

EQUAL ROOTS **Critically Damped Response** $\xi = 1$

$$p_1 = p_2 = -\frac{a_1}{2}, \text{ and } a_1^2 = 4a_2$$

For BIBO stability, $-\frac{a_1}{2} < 0 \square a_1 > 0$ **CONDITION #1**

$\square 4a_2 > 0 \square a_2 > 0$ **CONDITION #2**



BIBO Stability: Bottom Line

For all 3 scenarios, we found the same conditions for BIBO stability:

$$a_1 > 0, \text{ \& } a_2 > 0$$

ANY NEGATIVE COEFFICIENT(s) $\Rightarrow$ BIBO UNSTABLE



Example

For the following systems, by analyzing the coefficients alone (i.e., without solving the LCCDE), determine whether or not they are BIBO stable

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 3y(t) = 3x(t)$$
$$\frac{d^2 y(t)}{dt^2} - 2 \frac{dy(t)}{dt} + 3y(t) = x(t)$$
$$\frac{d^2 y(t)}{dt^2} + 8 \frac{dy(t)}{dt} + 3y(t) = 3x(t)$$



Impulse Response for Scenario C: A Problem!

When $p_1 = p_2$, our solution for $h(t)$ is invalid because the denominator becomes zero

$$h(t) = \frac{b_1 p_1 + b_2}{p_1 - p_2} e^{p_1 t} u(t) - \frac{b_1 p_2 + b_2}{p_1 - p_2} e^{p_2 t} u(t)$$

Alternate Strategy: We re-visit Table 2-2 for the case $p_1 = p_2$

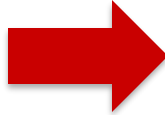


Table 2-2: Commonly encountered convolutions.

1. $u(t) * u(t) = t u(t)$

2. $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$

3. $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$

 **4.** $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$

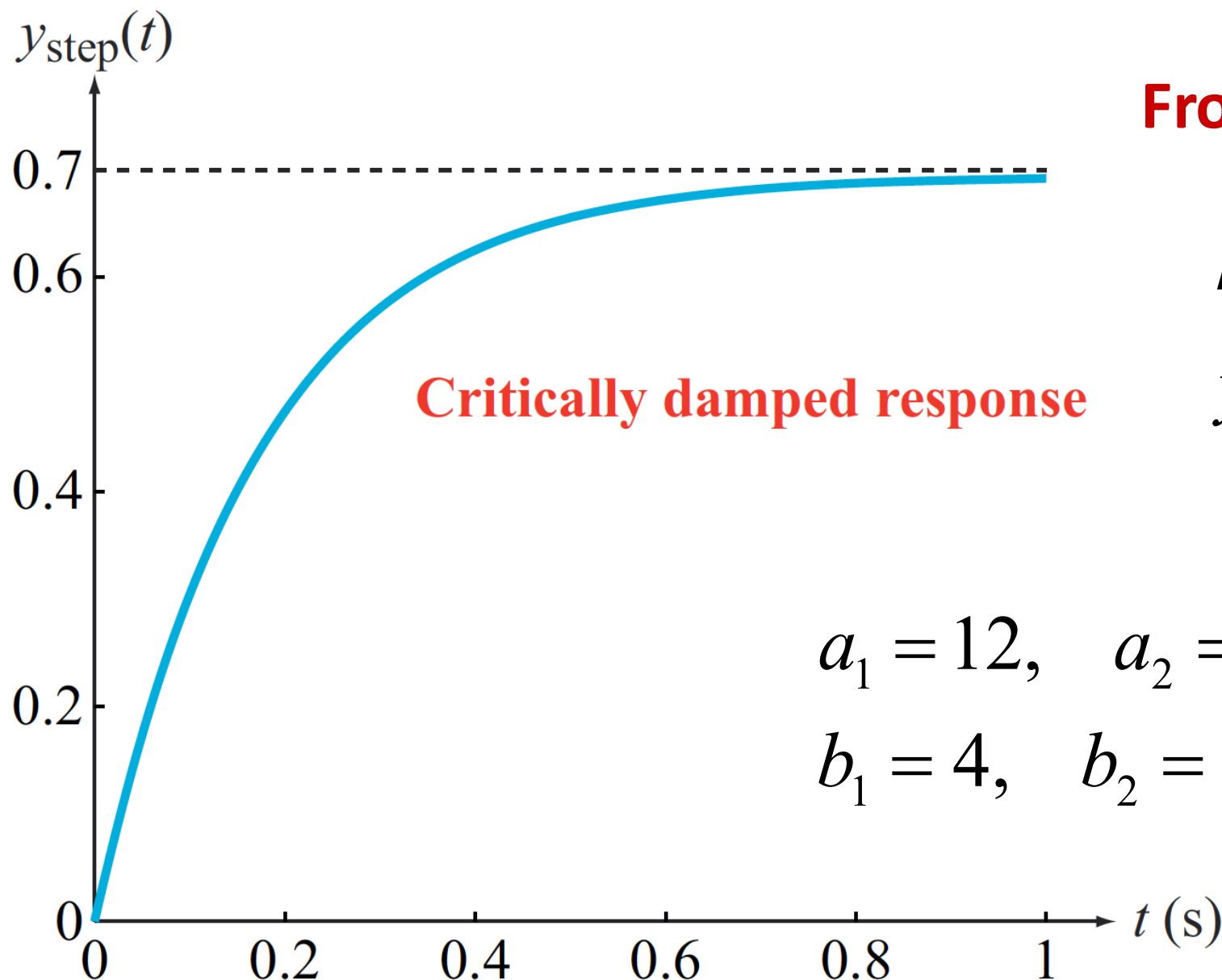
5. $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$
 $a \neq b$

6. $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$



Impulse Response for Scenario C:



From Table 2 – 2 Entry #4:

$$h_C(t) = te^{-\alpha t}u(t)$$

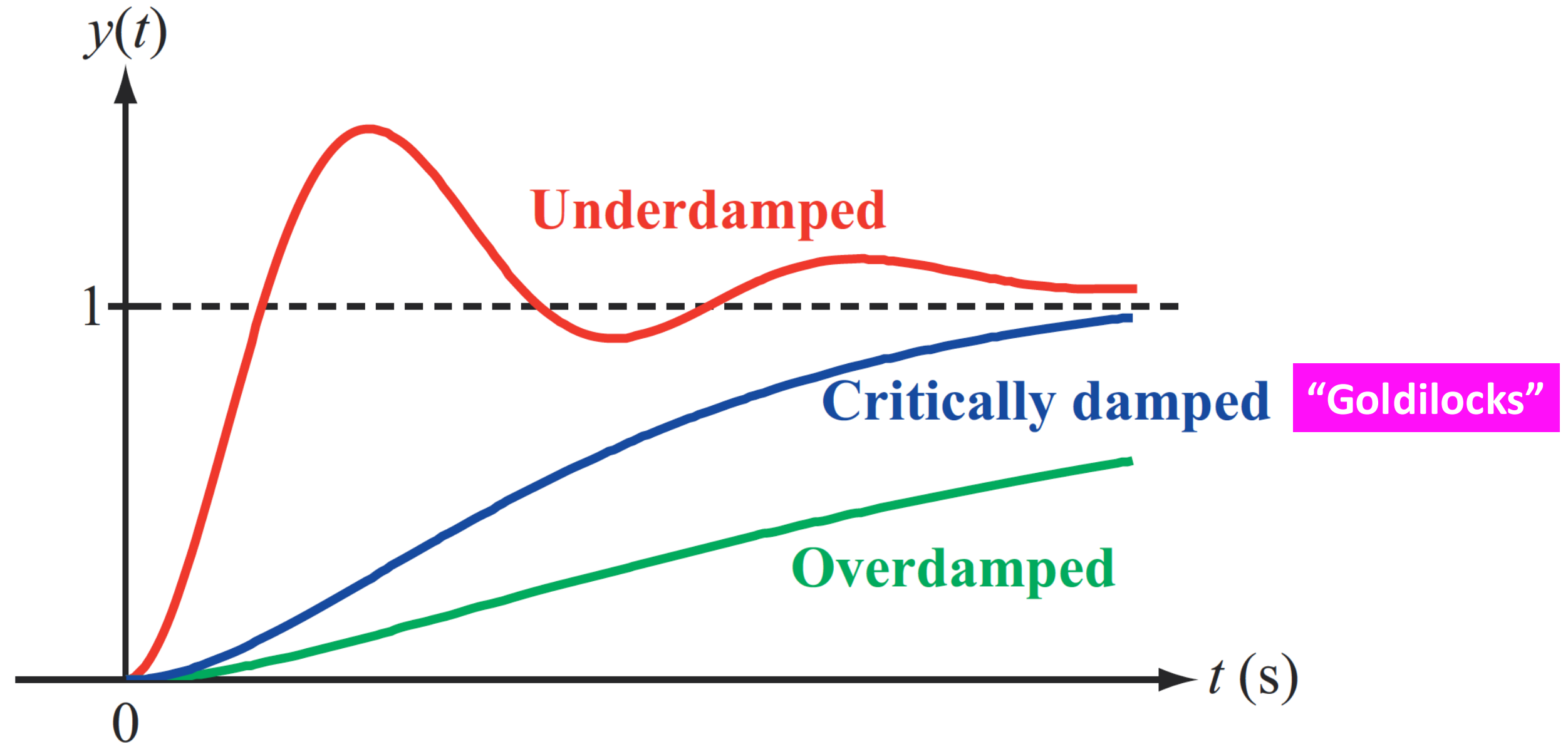
$$y_{\text{step}}(t) = h_C(t) * u(t)$$

$$a_1 = 12, \quad a_2 = 36$$

$$b_1 = 4, \quad b_2 = 25$$

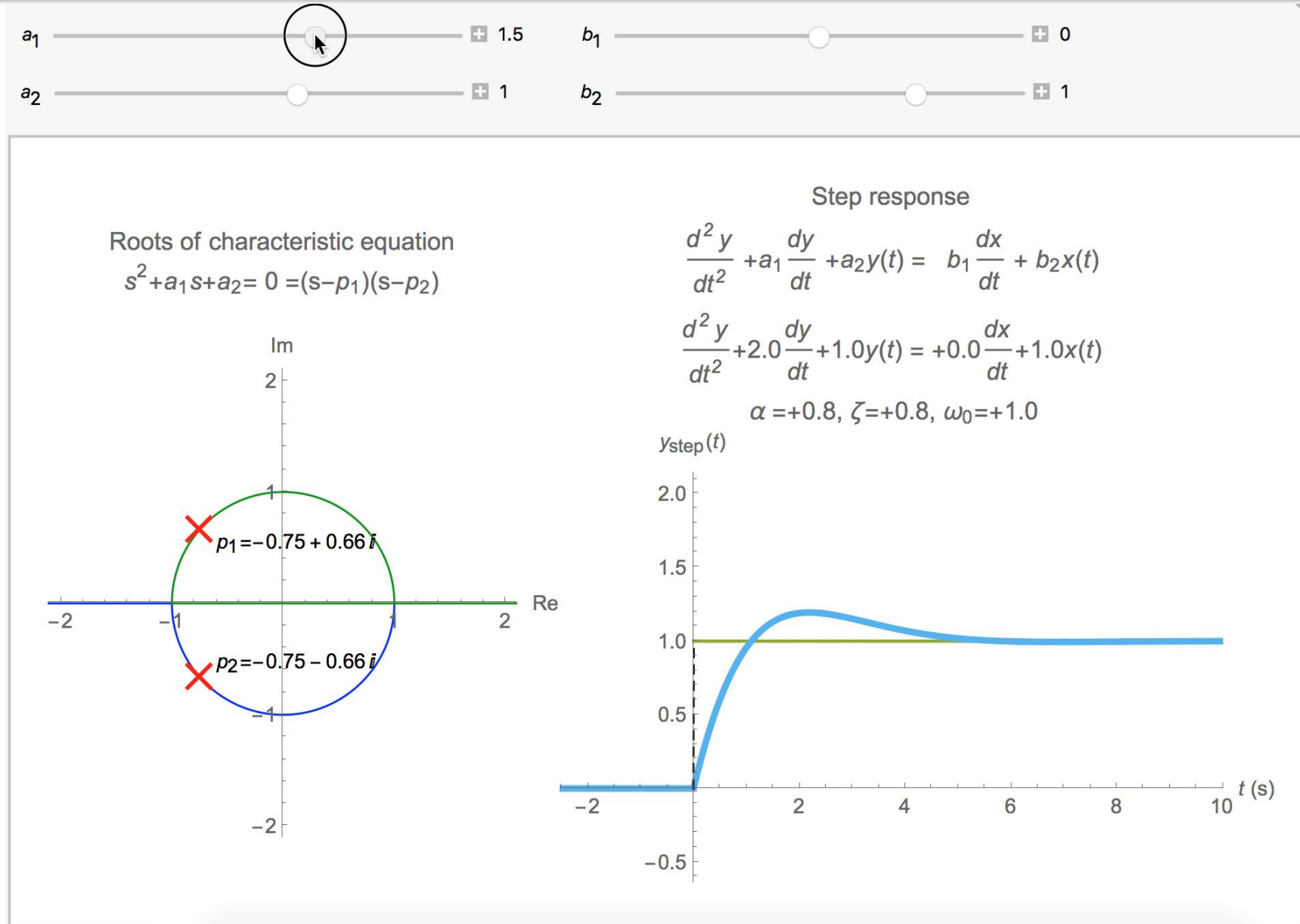


Summary





Video Summary of Today's Insights





Example

By examining the coefficients alone, determine whether each of these systems is BIBO stable, under-damped, over-damped, or critically damped

$$\begin{aligned}\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 3y(t) &= 3x(t) \\ \frac{d^2y(t)}{dt^2} - 2\frac{dy(t)}{dt} + 3y(t) &= x(t) \\ \frac{d^2y(t)}{dt^2} + 8\frac{dy(t)}{dt} + 3y(t) &= 3x(t)\end{aligned}$$



For the Next Class

Study Chapters 1 and 2 for Exam 1.

Old Exams with solutions are available for practice

Study Sections 3-1 and 3-2 for next class

How to Study for Exam 1:

- 1. Try the old exams without looking at the answers**
- 2. Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
- 3. Attempt the problems and practice until you can do them fully on your own without notes, like in a test**
- 4. For the test, speed is important, so practice for speed**
- 5. Write down your questions and bring them to the help sessions**